A compositional and diagrammatic syntax for parametric linear programming

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2026

Can optimization be an operation of the language itself?

One demand, two sources

The running example, used throughout the talk

$$\min\ x_1 + 2x_2 \quad \text{s.t.} \quad x_1 \geq 0,\ x_1 \leq 1,\ x_2 \geq 0,\ x_1 + x_2 \geq d.$$

Source 1: cost 1 per unit, capacity 1 $\implies$ decision x_1.

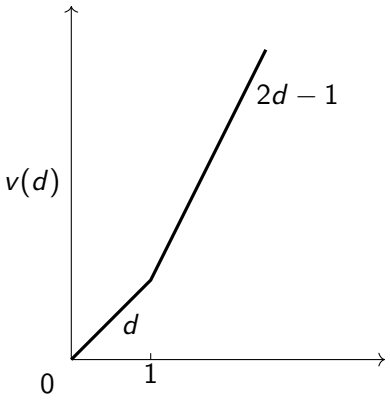
Source 2: cost 2 per unit, no capacity bound $\implies$ decision x_2.

Everything except the demand d is a fixed constant.

Takeaway

Varying the demand varies only a right-hand side parameter: one LP becomes a parametric family.

The value function is the observable



$$v(d) = \max\{0,\ d,\ 2d-1\}.$$

The value is piecewise affine and convex;

changing the formulation, adding variables, or splitting constraints leaves v unchanged.

The research question

Can LPs be composed and equated
by local diagrammatic rules?

Target of the dissertation

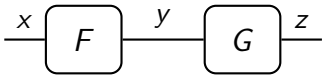
A sound, expressive, complete graphical language
for RHS-parametric linear programming.

Takeaway

Starting from GPA: **one new generator + three axioms**.

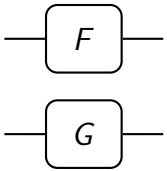
How to read the diagrams

Sequential composition



Shared wires are eliminated.

Parallel composition

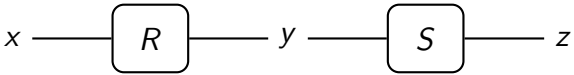


Independent blocks add their costs.

Takeaway

A wire carries a real variable; a box denotes a relation or cost bifunction.

Composition already eliminates variables

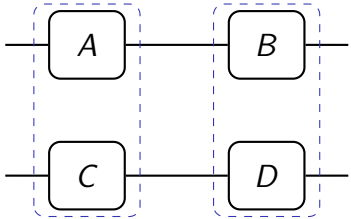


$$(S \circ R)(x, z) \iff \exists y : \; R(x, y) \wedge S(y, z).$$

Takeaway

The external interface retains x, z; the shared y disappears.

Reformulation can be a change of grouping



$$(B \circ A) \otimes (D \circ C) = (B \otimes D) \circ (A \otimes C).$$

Takeaway

The wiring is unchanged. Only the way we parse it changes.

Monoidal interchange law; dissertation, Introduction. Dashed boxes are grouping guides, not generators.

Replace feasibility by cost

$$\mathcal{I}_P(q) = \begin{cases} 0, & q \in P, \\ +\infty, & q \notin P. \end{cases}$$

Compose with these indicators and minimize over the shared variable:

Takeaway

Existential elimination becomes optimal aggregation:

$$(G \circ F)(x, z) = \inf_y \big(F(x, y) + G(y, z)\big).$$

For costs in $\{0, +\infty\}$ this is exactly feasibility of the connection.

Quantales explain the shared pattern

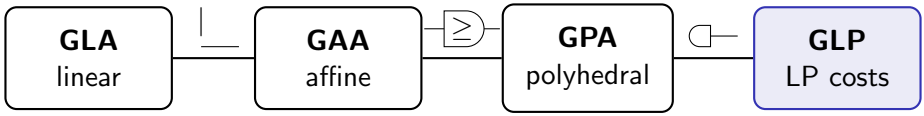
$$(S \circ R)(x, z) = \bigvee_{y} R(x, y) \otimes S(y, z).$$

Setting	**Combine**	**Aggregate**
Truth values	$\wedge$	$\exists$ / $\bigvee$
Costs	$+$	inf

Takeaway

Quantale: a complete lattice with a join-preserving monoidal product.

Build on the existing graphical languages



Takeaway

Each extension adds expressive power.

GPA expresses feasible regions



$$[\![\geq]\!] = \{(u, v) \mid u \geq v\}.$$

Matrices + affine constants + order

$$P = \{q \mid Mq + r \geq 0\}.$$

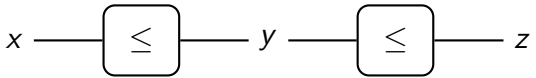
Equalities are pairs of inequalities.

Takeaway

At this stage every feasible point costs 0; every infeasible point costs $+\infty$.

Projection is the computational primitive

$$\exists y: \quad y \geq x, \quad z \geq y \qquad \Longleftrightarrow \qquad z \geq x.$$



Takeaway

Fourier–Motzkin elimination pairs lower bounds with upper bounds.

GPA provides the proof engine

Existing result, developed in the draft

$$\mathrm{FreeP}_{\mathbf{GPA}} \cong \textbf{PolyRel}.$$

Expressivity

Every polyhedral relation has a diagram.

Soundness

Every allowed rewrite preserves its relation.

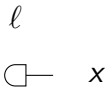
Completeness

Equal relations are equated by the rules.

Takeaway

Later, LP equality will be transferred to equality of polyhedral relations.

The new generator assigns linear cost



$$[\![\ell]\!](\bullet, x) = x.$$

The output is a variable, not a chosen optimizer.

$\ell : 0 \longrightarrow 1$

Composition with constraints will optimize over it.

Takeaway

One scalar cost suffices for arbitrary linear objectives.

The semantics: LPs with external interfaces

A morphism $F : n \to m$ is represented by a value function:

$$F(x, y) = \inf_{z} \left\{ c^{\top} z \;\middle|\; Az \geq B \begin{bmatrix} x \\ y \end{bmatrix} + r \right\}.$$

x, y: external variables. z: internal decisions.

A, B, c, r are fixed; the RHS depends affinely on the interface.

Takeaway

The denotation is a value function, not an optimization algorithm.

LP blocks compose into LP blocks

$$(G \circ F)(x, w) = \inf_{y,z,z'} \left(c_F^\top z + c_G^\top z' \right)$$

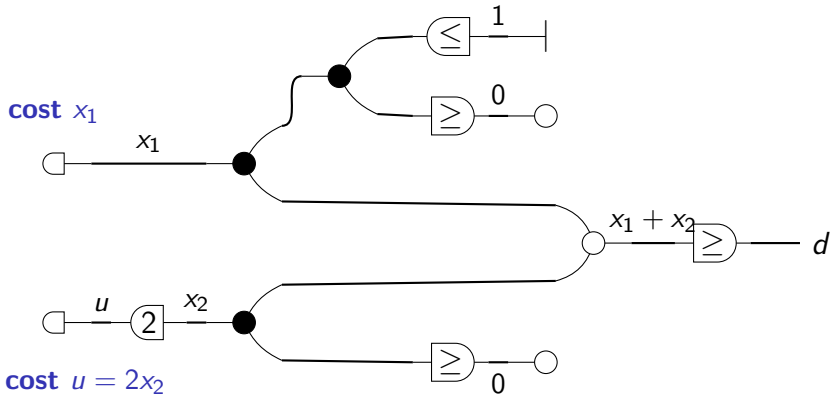
subject to both blocks' linear constraints.

$$(F \otimes G)\big((x, x'), (y, y')\big) = F(x, y) + G(x', y').$$

Takeaway

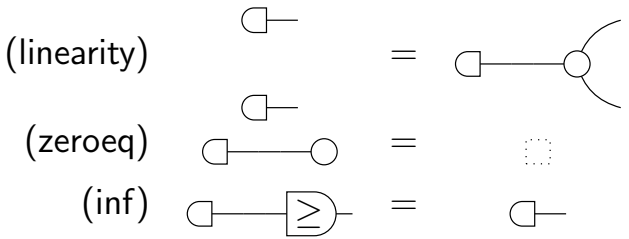
Sequential composition eliminates; parallel composition adds costs.

Our running LP becomes a diagram



$$v(d) = \inf\{x_1 + 2x_2 \mid x_1 + x_2 \geq d,\ 0 \leq x_1 \leq 1,\ x_2 \geq 0\}.$$

Three axioms control the new cost



Their semantic content

Additivity: $\ell(x) + \ell(y) = \ell(x + y)$.

Zero: evaluate the cost at 0.

Infimum: $\inf_{t \geq x} t = x$.

Takeaway

The third axiom is the link between a cost and its epigraph.

Expressivity and soundness are direct

Expressivity

1. Encode the linear objective.
2. Encode constraints in GPA.
3. Compose to eliminate decisions.

Soundness

Inherited GPA rules are sound.

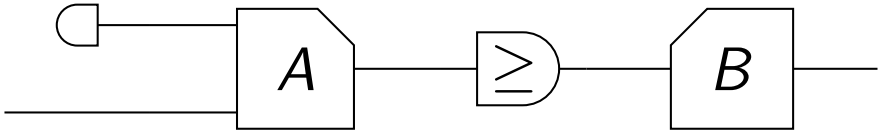
Check the three new identities.

Rewrites preserve denotation.

Takeaway

The difficult direction is the converse: can the rules prove *every* value-function equality?

Normal form collects the objective



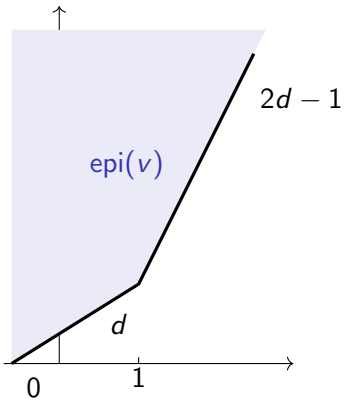
One cost source + a polyhedral constraint block

Structural induction on generators, sequential composition, and tensor.

Takeaway

A common *shape* for diagrams, not a claim of unique matrices or a unique LP formulation.

The epigraph turns cost into geometry



$$\mathrm{epi}(v) = \{(d, t) \mid t \geq v(d)\}.$$

For our example the shaded region is exactly

$$t \geq 0, \qquad t \geq d, \qquad t \geq 2d - 1,$$

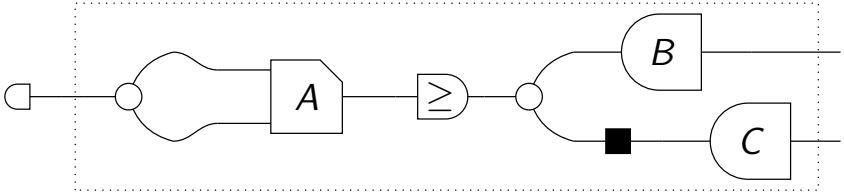
a polyhedron, hence a GPA diagram.

Takeaway

Same value function⇒same epigraph: an equality of costs becomes an equality of polyhedra.

Epigraph form is a syntactic bridge

Adding a threshold variable, the epigraph of a value function is a GPA relation E, and $F(q) = \inf\{t \mid (q,t) \in E\}$:

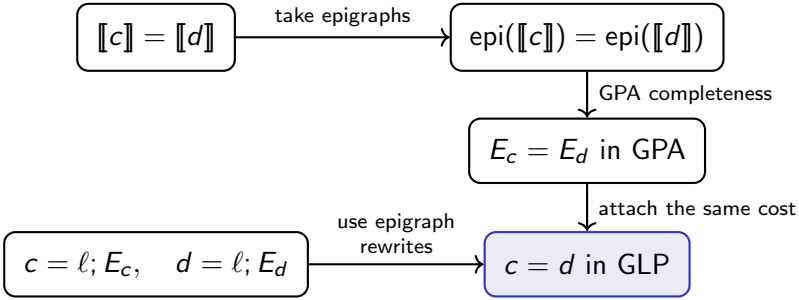


$$\mathrm{epi}([d]) = \{(t, y_1, y_2) \mid \exists z \le t :\ Az + a \ge By_1 - Cy_2\}, \qquad A = [A'\ C].$$
$$[d](y_1, y_2) = \inf_z \{z \mid Az + a \ge By_1 - Cy_2\}.$$

Takeaway

Crucially, the axioms *rewrite* the diagram into this: one cost source, one GPA block.

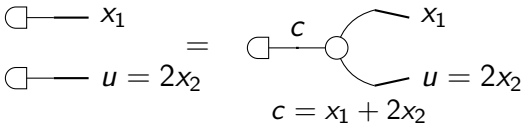
The completeness argument



Semantic equality yields an equational derivation.

Simplify the example: collect cost, then eliminate

1. Linearity collects the two cost sources.



2. The (inf) rule introduces $t \geq c$; GPA projects the decisions.

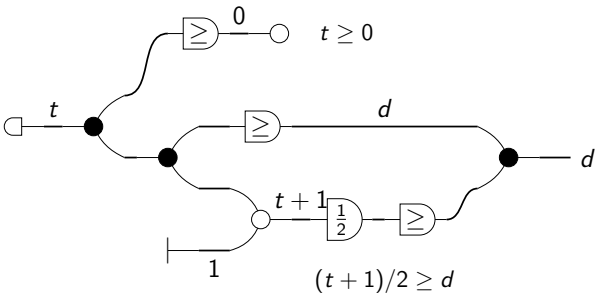
$$s = x_1 + x_2, \quad c = 2s - x_1 : \qquad \max\{0, 2s - t\} \leq x_1 \leq \min\{1, s\}.$$

$$\xLeftrightarrow[\text{GPA / Fourier–Motzkin: eliminate } x_1]{} \quad s \geq 0, \quad t \geq s, \quad t \geq 2s - 1, \quad s \geq d.$$

$$\xLeftrightarrow[\text{GPA / Fourier–Motzkin: eliminate } s]{} \quad \boxed{t \geq 0, \qquad t \geq d, \qquad t \geq 2d - 1.}$$

Each projection is an inherited GPA rewrite; redundant bounds are removed.

The simplified diagram has no source decisions



$$v(d) = \inf\{t \mid t \geq 0,\ t \geq d,\ t \geq 2d-1\} = \max\{0, d, 2d-1\}.$$

Linearity + (inf) + GPA: two decisions become one cost wire.

Result, scope, and remaining work

$$\mathrm{FreeP}_{\mathbf{GLP}} \cong \mathbf{LPRel}$$

Presented in this work

RHS-parametric value functions.

One generator; three axioms.

Normal form and completeness.

Optimization becomes part of the syntax

Composition eliminates internal decisions.

Equations certify equality of value functions.

Thank you. Questions?